

Topology PhD Exam, August 2020

Work the following problems and show all work. Support all statements to the best of your ability.

- Let X be a topological space. Given two points $a, b \in X$, a collection of sets $A_1, \dots, A_n$ is a simple chain from a to b if $a \in A_1$, $b \in A_n$, and $A_i \cap A_j \neq \emptyset$ if and only if $|i - j| \leq 1$. Prove that if $\{U_\alpha\}$ is a collection of open sets covering X and X is connected, then there is a simple chain of elements in $\{U_\alpha\}$ joining any pair of points $a, b \in X$. (Hint: Consider the set C_a of all b so that there is a simple chain of elements joining a and b . Show that C_a is both open and closed.)
- Let $X = S^2 \cup_f e^3 \cup_h e^5$, where $f : S^2 \rightarrow S^2$ has degree n and $h : S^4 \rightarrow S^2 \cup_f e^3$ is continuous.
 - What is the Euler characteristic of X ?
 - Determine $H_\bullet(X; \mathbb{Z})$ and $H^\bullet(X; \mathbb{Z}_2)$ as a function of n .
 - Can X have the homotopy type of a closed 5-manifold? (If the answer depends on n , explain.)
- Let X be the space $\mathbb{RP}^3/\mathbb{RP}^1$ (the space obtained by collapsing $\mathbb{RP}^1 \subset \mathbb{RP}^3$ to a point). Compute the integral homology groups of X .
- Let $\mathbb{RP}^2$ be the real projective plane and fix a basepoint $x_0 \in \mathbb{RP}^2$. Note that we can realize the wedge sum $\mathbb{RP}^2 \vee \mathbb{RP}^2$ as $(\mathbb{RP}^2 \times \{x_0\}) \cup (\{x_0\} \times \mathbb{RP}^2) \subset \mathbb{RP}^2 \times \mathbb{RP}^2$.
 - Compute the fundamental groups of $\mathbb{RP}^2 \vee \mathbb{RP}^2$ and $\mathbb{RP}^2 \times \mathbb{RP}^2$.
 - Show that $\mathbb{RP}^2 \vee \mathbb{RP}^2$ is not a retract of $\mathbb{RP}^2 \times \mathbb{RP}^2$.
 - Show that any map $f : \mathbb{RP}^2 \vee \mathbb{RP}^2 \rightarrow S^1$ is null-homotopic.
- Compute $H_\bullet(M; \mathbb{Z}_2)$, where M is the space with integral homology groups

$$H_k(M; \mathbb{Z}) = \begin{cases} \mathbb{Z} & k = 0, 2, 4, \\ \mathbb{Z}_2 & k = 1, \\ \mathbb{Z}_3 & k = 3, \\ 0 & \text{otherwise.} \end{cases}$$

Answer the following with complete definitions, statements, or short proofs.

- Let M be an orientable simply-connected 3-manifold. Compute the integral homology of M .
- Compute $\chi(\mathbb{RP}^2 \times \mathbb{CP}^4 \times S^4)$.
- Give an example of a space that is connected but not locally connected.
- State the Urysohn Lemma.
- Prove that if $m \neq n$, then $\mathbb{R}^m$ is not homeomorphic to $\mathbb{R}^n$.
- Let $\mathbb{N}$ be the set of natural numbers and let $\mathbb{N}^*$ be its one-point compactification. Let X be any topological space. Prove that a sequence $\{a_n\}$ in X converges to a if and only if the map $f : \mathbb{N}^* \rightarrow X$ defined by $f(n) = a_n$ and $f(\infty) = a$ is continuous. (Here $\mathbb{N}$ has the discrete topology.)

12. State the Borsuk–Ulam Theorem.

13. Describe the possible connected covering spaces of $S^1 \times S^1$.

14. Does the following exact sequence of abelian groups necessarily split? Prove or give a counterexample.

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \oplus \mathbb{Z} \rightarrow A \rightarrow 0$$

15. Compute the integral homology of the space $\mathbb{CP}^2 \times \mathbb{RP}^3$.